BSc 2024/2025

For internal students of Imperial College.

Mathematics Foundations (BUSI40001)

3 hours

XXXday ABth January 2024, 09:00 – 12:00

Instructions for the exam

- For Section A, internet access with ChatGPT is allowed and a computer with R or Python is needed.
- Section B is closed-book and paper-based. Section B is to made accessible to the students after completing Section A.
- Answer ALL questions (The mark breakdown is indicated throughout the exam).
- It is important to write neatly and clearly.
- If a question or part of a question is unclear to you, state any assumptions you need in order to complete the question.

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SECTION A (35 marks)

You have 45 minutes for Section A. Answer all **TWO** questions. Please submit your code. You must close your laptop after 45 minutes.

1. (20 marks) The ages at death of male parents and their male children are shown in the table below.

Person	A	B	C	D	E	F	G
adult	75	74	85	40	69	83	65
child	84	82	93	75	77	90	73

This question is about a male's lifespan given his father's lifespan.

(a) Write out the linear regression in the matrix form

$$\underline{y} = \mathbf{X}\underline{\beta} + \underline{\epsilon},$$

where $\underline{y}$, $\mathbf{X}$, $\underline{\beta}$, and $\underline{\epsilon}$ are to be defined. [2 marks]

[1/2 a mark for each vector]

$$\underline{x} = (75, 74, 85, 40, 69, 83, 65)^\top$$

is the 7×1 vector of male parent lifespans,

$$\underline{y} = (84, 82, 93, 75, 77, 90, 73)^\top$$

is the 7×1 vector of male child lifespans, $\underline{e} = (1, \ldots, 1)^\top$ is 7×1 vector of ones.

$$\mathbf{X} = [\underline{e}, \underline{x}]$$

is the 7 by 2 design matrix,

$$\underline{\beta} = [\beta_1, \beta_2]^\top$$

is the 1 by 2 parameter vector, and $\underline{\epsilon}$ is 7×1 error vector.

(b) Show that the expression for $\underline{\beta}$ which minimizes $\underline{\epsilon}^\top \underline{\epsilon}$ is given by

$$\widehat{\underline{\beta}} = \left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\underline{y}.$$

[5 marks]

Observe that $\underline{\epsilon}^\top \underline{\epsilon}$, where $\underline{\epsilon} = \underline{y} - \mathbf{X}\underline{\beta}$, is minimized by projecting $\underline{y}$ orthogonally into the column space of $\mathbf{X}$. Therefore, we want to choose $\underline{\beta}$ such that $\mathbf{X}^\top(\underline{y} - \mathbf{X}\underline{\beta}) = \underline{0}$. We shall denote the parameter vector which satisfies the latter equation via $\widehat{\underline{\beta}}$ and so

$$\mathbf{X}^\top(\underline{y} - \mathbf{X}\widehat{\underline{\beta}}) = \underline{0}$$

Therefore,

$$\mathbf{X}^\top\underline{y} = \mathbf{X}^\top\mathbf{X}\widehat{\underline{\beta}}.$$

Now $\mathbf{X}^\top\mathbf{X}$ is a 2 by 2 square matrix. Now

$$\widehat{\underline{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\underline{y}.$$

(c) Show that the fitted values are given by

$$\underline{\hat{y}} = \mathbf{H}\underline{y}$$

where

$$\mathbf{H} \equiv \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T$$

[3 marks]

The fitted values of the dependent variable are given by $\underline{\hat{y}} = \underline{\widehat{\beta}}\mathbf{X}$. Therefore,

$$\underline{\hat{y}} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\underline{y},$$

and so

$$\underline{\hat{y}} = \mathbf{H}\underline{y},$$

where

$$\mathbf{H} \equiv \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T.$$

(d) Show that $\mathbf{H}$ is idempotent and symmetric. [5 marks]

$$\begin{aligned}
\mathbf{H}^2 &= \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T \\
&= \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T \\
&= \mathbf{H},
\end{aligned}$$

and so $\mathbf{H}$ is idempotent. Now,

$$\begin{aligned}
\mathbf{H}^\top &= [\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T]^\top \\
&= [\mathbf{X}^\top]^\top[\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}]^\top \\
&= \mathbf{X}[\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}]^\top \\
&= \mathbf{X}[\left(\mathbf{X}^T\mathbf{X}\right)^{-1}]^\top\mathbf{X}^\top \\
&= \mathbf{X}[\left(\mathbf{X}^T\mathbf{X}\right)^\top]^{-1}\mathbf{X}^\top \\
&= \mathbf{X}\left[\left(\mathbf{X}^\top(\mathbf{X}^T)^\top\right)\right]^{-1}\mathbf{X}^\top \\
&= \mathbf{X}\left[\left(\mathbf{X}^\top\mathbf{X}\right)\right]^{-1}\mathbf{X}^\top \\
&= \mathbf{H},
\end{aligned}$$

and so $\mathbf{H}$ is symmetric.

(e) Estimate the lifetime of a male whose male parent dies at age 89. You should do this using either R or Python [5 marks]

You need a calculator or a computer!

$$\begin{aligned}
\mathbf{X}^\top\mathbf{X} &= \begin{pmatrix} 7 & 491 \\ 491 & 35801 \end{pmatrix} \\
\left[\left(\mathbf{X}^\top\mathbf{X}\right)\right]^{-1} &= \begin{pmatrix} \frac{35801}{9526} & -\frac{491}{9526} \\ -\frac{491}{9526} & \frac{7}{9526} \end{pmatrix} \\
\left[\left(\mathbf{X}^\top\mathbf{X}\right)\right]^{-1}\mathbf{X}^\top &= \begin{pmatrix} -\frac{512}{4763} & -\frac{533}{9526} & -\frac{2967}{4763} & \frac{16161}{9526} & \frac{961}{4763} & -\frac{2476}{4763} & \frac{1943}{4763} \\ \frac{17}{4763} & \frac{27}{9526} & \frac{52}{4763} & -\frac{211}{9526} & -\frac{4}{4763} & \frac{45}{4763} & -\frac{18}{4763} \end{pmatrix} \\
\underline{\widehat{\beta}} &= \left[\left(\mathbf{X}^\top\mathbf{X}\right)\right]^{-1}\mathbf{X}^\top\underline{y} \\
&= \begin{pmatrix} \frac{46953}{866} \\ \frac{343}{866} \end{pmatrix} \\
&= \begin{pmatrix} 54.2182 \\ 0.396074 \end{pmatrix}
\end{aligned}$$

$$54.2182 + 0.396074 \times 89 = 89.4688$$

Estimate of age at death for male child is 89.4688.

2. (15 marks: 5 marks for explanation, 10 marks for code)

Find and classify the critical points of the function $f(x, y) = x^2 + y^3 + xy$. Use code written in R or Python to carry out the necessary calculations, but explain the steps needed without showing algebraic details.

To find the critical points, we first compute the first-order partial derivatives of f with respect to x and y, set them equal to zero, and solve for x and y.

$$\frac{\partial f}{\partial x} = 0$$
$$\frac{\partial f}{\partial y} = 0$$

[1 mark]

So one critical point is at $(0, 0)$.

[1 mark]

The other critical point is at $\left(-\frac{1}{12}, \frac{1}{6}\right)$.

[1 mark]

We compute the second-order partial derivatives and use them to find the determinant of the Hessian.

The determinant D of the Hessian matrix is given by:

$$D = f_{xx}f_{yy} - (f_{xy})^2$$

- At $(0, 0)$: $D = -1$. Since $D < 0$, the critical point at $(0, 0)$ is a saddle point.

[1 mark]

At $\left(-\frac{1}{12}, \frac{1}{6}\right)$, $D > 0$ and $f_{xx} > 0$, the critical point is a local minimum.

[2 marks]

```
import sympy as sp

# Define symbols
x, y = sp.symbols('x y')

# Define the function
f = x**2 + y**3 + x*y

# Compute first-order partial derivatives
df_dx = sp.diff(f, x)
df_dy = sp.diff(f, y)

# Solve the system of equations df/dx = 0 and df/dy = 0
critical_points = sp.solve([df_dx, df_dy], (x, y))

# Compute second-order partial derivatives
f_xx = sp.diff(df_dx, x)
f_yy = sp.diff(df_dy, y)
f_xy = sp.diff(df_dx, y)

# Function to classify critical points
def classify_critical_point(point):
    x0, y0 = point
    D = f_xx.subs({x: x0, y: y0}) * f_yy.subs({x: x0, y: y0}) -
    f_xy.subs({x: x0, y: y0})**2
    f_xx_val = f_xx.subs({x: x0, y: y0})
    if D > 0:
        if f_xx_val > 0:
            classification = 'Local minimum'
        else:
            classification = 'Local maximum'
    elif D < 0:
        classification = 'Saddle point'
    else:
        classification = 'Indeterminate'
    return classification

# Output the results
for point in critical_points:
    classification = classify_critical_point(point)
    print(f"Critical point at {point}: {classification}")


Critical point at (0, 0): Saddle point
Critical point at (-1/12, 1/6): Local minimum
```

SECTION B (80 marks)

You have 2 hours 15 minutes for Section B. Answer all **FOUR** questions. Each question is worth 20 marks.

Question 1 (20 marks)

(a) Define what is meant by a *subsequence* of a sequence (a_n). [3 marks]

(b) Prove that if a sequence (a_n) converges to a limit L, then every subsequence also converges to L. [6 marks]

(c) Prove that every bounded sequence of real numbers has a monotone subsequence. [6 marks]

(d) State the Bolzano-Weierstrass Theorem. [2 marks]

(e) What is meant by the *peak* of a sequence? [3 marks]

(a)
A *subsequence* of a sequence (a_n) is a sequence (a_{n_k}) where (n_k) is a strictly increasing sequence of positive integers. In other words, $n_1 < n_2 < n_3 < \cdots$, and the subsequence consists of the terms $a_{n_1}, a_{n_2}, a_{n_3}, \ldots$ selected from the original sequence while preserving their order.

(b) Let $(a_n) \to L$ and let (a_{n_k}) be any subsequence of (a_n).
Since $(a_n) \to L$, for any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, we have $|a_n - L| < \varepsilon$.
Since (n_k) is strictly increasing and $n_k \geq k$ for all $k \in \mathbb{N}$, there exists $K \in \mathbb{N}$ such that $n_K \geq N$.
For all $k \geq K$, we have $n_k \geq n_K \geq N$, and therefore $|a_{n_k} - L| < \varepsilon$.
This shows that $(a_{n_k}) \to L$, completing the proof.

(c) Let (a_n) be a bounded sequence. Define a term a_n to be a *peak* if $a_n \geq a_m$ for all $m > n$.
Case 1: The sequence has infinitely many peaks. Let $a_{n_1}, a_{n_2}, a_{n_3}, \ldots$ be the peaks in order. By definition, $a_{n_1} \geq a_{n_2} \geq a_{n_3} \geq \cdots$. Thus (a_{n_k}) is a monotone decreasing subsequence.
Case 2: The sequence has finitely many peaks.
Let N be such that there are no peaks after index N. For $n_1 > N$, since a_{n_1} is not a peak, there exists $n_2 > n_1$ such that $a_{n_2} > a_{n_1}$. Similarly, since a_{n_2} is not a peak, there exists $n_3 > n_2$ such that $a_{n_3} > a_{n_2}$.
Continuing this process, we obtain a strictly increasing subsequence (a_{n_k}).
In both cases, we have found a monotone subsequence.

(d) Bolzano-Weierstrass Theorem: Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.

(e) A term a_n of a sequence (a_n) is called a *peak* if $a_n \geq a_m$ for all $m > n$. In other words, a_n is a peak if it is greater than or equal to all subsequent terms in the sequence.

Question 2 (20 marks)

(a) Define a *vector subspace* and give two examples in $\mathbb{R}^3$. [4 marks]

(b) Let U and W be subspaces of a vector space V. Define $U \cap W$ and $U + W$, and prove that both are subspaces of V. [6 marks]

(c) Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear transformation defined by $T(x, y) = (x + 2y, 3x - y)$. Find the matrix representation of T and determine its eigenvalues. [6 marks]

(d) Is T invertible? Justify your answer. [4 marks]

(a) A *vector subspace* W of a vector space V is a non-empty subset of V that is closed under vector addition and scalar multiplication. That is:

- $\mathbf{0} \in W$
- If $\mathbf{u}, \mathbf{v} \in W$, then $\mathbf{u} + \mathbf{v} \in W$
- If $\mathbf{u} \in W$ and $c \in \mathbb{R}$, then $c\mathbf{u} \in W$

Two examples in $\mathbb{R}^3$:

- The plane through the origin: $W_1 = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$
- The line through the origin: $W_2 = \{(t, 2t, -t) : t \in \mathbb{R}\}$

(b) Let U and W be subspaces of V.
$U \cap W = \{\mathbf{v} \in V : \mathbf{v} \in U \text{ and } \mathbf{v} \in W\}$
$U + W = \{\mathbf{u} + \mathbf{w} : \mathbf{u} \in U, \mathbf{w} \in W\}$
Proof that $U \cap W$ is a subspace:

- Since U and W are subspaces, $\mathbf{0} \in U$ and $\mathbf{0} \in W$, so $\mathbf{0} \in U \cap W$.
- If $\mathbf{v}_1, \mathbf{v}_2 \in U \cap W$, then $\mathbf{v}_1, \mathbf{v}_2 \in U$ and $\mathbf{v}_1, \mathbf{v}_2 \in W$. Since U and W are subspaces, $\mathbf{v}_1 + \mathbf{v}_2 \in U$ and $\mathbf{v}_1 + \mathbf{v}_2 \in W$. Therefore, $\mathbf{v}_1 + \mathbf{v}_2 \in U \cap W$.
- If $\mathbf{v} \in U \cap W$ and $c \in \mathbb{R}$, then $\mathbf{v} \in U$ and $\mathbf{v} \in W$. Since U and W are subspaces, $c\mathbf{v} \in U$ and $c\mathbf{v} \in W$. Therefore, $c\mathbf{v} \in U \cap W$.

Proof that $U + W$ is a subspace:

- $\mathbf{0} = \mathbf{0} + \mathbf{0} \in U + W$ since $\mathbf{0} \in U$ and $\mathbf{0} \in W$.
- If $\mathbf{v}_1 = \mathbf{u}_1 + \mathbf{w}_1$ and $\mathbf{v}_2 = \mathbf{u}_2 + \mathbf{w}_2$ with $\mathbf{u}_1, \mathbf{u}_2 \in U$ and $\mathbf{w}_1, \mathbf{w}_2 \in W$, then
$$\mathbf{v}_1 + \mathbf{v}_2 = (\mathbf{u}_1 + \mathbf{u}_2) + (\mathbf{w}_1 + \mathbf{w}_2) \in U + W$$
since $\mathbf{u}_1 + \mathbf{u}_2 \in U$ and $\mathbf{w}_1 + \mathbf{w}_2 \in W$.
- If $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in U$, $\mathbf{w} \in W$, and $c \in \mathbb{R}$, then $c\mathbf{v} = c\mathbf{u} + c\mathbf{w} \in U + W$ since $c\mathbf{u} \in U$ and $c\mathbf{w} \in W$.

(c) The matrix representation of T with respect to the standard basis is found by applying T to the basis vectors:
$$T(1, 0) = (1, 3) \quad \text{and} \quad T(0, 1) = (2, -1)$$
Therefore, the matrix representation is:
$$A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$$

To find eigenvalues, we solve $\det(A - \lambda I) = 0$:

$$\det \begin{pmatrix} 1-\lambda & 2 \\ 3 & -1-\lambda \end{pmatrix} = (1-\lambda)(-1-\lambda) - 6 = -1 - \lambda + \lambda + \lambda^2 - 6 = \lambda^2 - 7$$

Setting $\lambda^2 - 7 = 0$, we get $\lambda = \pm\sqrt{7}$.

Eigenvector for $\lambda_1 = \sqrt{7}$: Solve $(A - \sqrt{7}I)\mathbf{v} = \mathbf{0}$:

$$\begin{pmatrix} 1-\sqrt{7} & 2 \\ 3 & -1-\sqrt{7} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

From the first equation: $(1-\sqrt{7})x + 2y = 0 \Rightarrow y = \frac{\sqrt{7}-1}{2}x$

Choosing $x = 2$ gives $\mathbf{v}_1 = \begin{pmatrix} 2 \\ \sqrt{7}-1 \end{pmatrix}$

Eigenvector for $\lambda_2 = -\sqrt{7}$: Solve $(A + \sqrt{7}I)\mathbf{v} = \mathbf{0}$:

$$\begin{pmatrix} 1+\sqrt{7} & 2 \\ 3 & -1+\sqrt{7} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

From the first equation: $(1+\sqrt{7})x + 2y = 0 \Rightarrow y = -\frac{1+\sqrt{7}}{2}x$

Choosing $x = 2$ gives $\mathbf{v}_2 = \begin{pmatrix} 2 \\ -1-\sqrt{7} \end{pmatrix}$

(d) T is invertible if and only if $\det(A) \neq 0$.
$\det(A) = (1)(-1) - (2)(3) = -1 - 6 = -7 \neq 0$.
Therefore, T is invertible.
Alternatively, since both eigenvalues $\sqrt{7}$ and $-\sqrt{7}$ are non-zero, the transformation has no non-trivial kernel, confirming that T is invertible.

Question 3 (20 marks)

(a) Let A be the 3×3 matrix given by:

$$A = \begin{pmatrix} a & b & 0 \\ b & a & 0 \\ 0 & 0 & c \end{pmatrix},$$

where $a, b, c \in \mathbb{R}$ and $c \neq 0$. Find the characteristic polynomial of A. [5 marks]

(b) Determine the eigenvalues of A in terms of a, b, and c. [5 marks]

(c) For the case where $b \neq 0$, find the eigenvectors corresponding to each eigenvalue. [5 marks]

(d) Diagonalize the matrix A if possible, and explain your reasoning. [5 marks] (**not covered in 2025/2026**)

(a) The characteristic polynomial is $\det(A - \lambda I)$:

$$\det \begin{pmatrix} a-\lambda & b & 0 \\ b & a-\lambda & 0 \\ 0 & 0 & c-\lambda \end{pmatrix}$$

Using expansion along the third row:

$$= (c-\lambda)\det\begin{pmatrix} a-\lambda & b \\ b & a-\lambda \end{pmatrix} = (c-\lambda)[(a-\lambda)^2 - b^2]$$

Therefore, the characteristic polynomial is:

$$p(\lambda) = (c-\lambda)[(a-\lambda)^2 - b^2] = (c-\lambda)(\lambda^2 - 2a\lambda + a^2 - b^2)$$

(b) Setting the characteristic polynomial equal to zero:

$$(c-\lambda)[(a-\lambda)^2 - b^2] = 0$$

This gives us:

- $\lambda_1 = c$
- $(a-\lambda)^2 - b^2 = 0 \Rightarrow (a-\lambda)^2 = b^2 \Rightarrow a - \lambda = \pm b$
- $\lambda_2 = a - b$
- $\lambda_3 = a + b$

Therefore, the eigenvalues are: $\lambda_1 = c$, $\lambda_2 = a - b$, and $\lambda_3 = a + b$.

(c) For $b \neq 0$:

For $\lambda_1 = c$: Solve $(A - cI)\mathbf{v} = \mathbf{0}$:

$$\begin{pmatrix} a-c & b & 0 \\ b & a-c & 0 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

This gives $(a-c)x + by = 0$ and $bx + (a-c)y = 0$. If $a \neq c$, these equations force $x = y = 0$, and z is free. Thus $\mathbf{v}_1 = (0,0,1)^T$.

For $\lambda_2 = a - b$: Solve $(A - (a-b)I)\mathbf{v} = \mathbf{0}$:

$$\begin{pmatrix} b & b & 0 \\ b & b & 0 \\ 0 & 0 & c-a+b \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

This gives $b(x+y) = 0$, so $x + y = 0$ (since $b \neq 0$). Also $(c-a+b)z = 0$. If $c \neq a - b$, then $z = 0$. Choosing $x = 1$ gives $y = -1$. Thus $\mathbf{v}_2 = (1,-1,0)^T$.

For $\lambda_3 = a + b$: Solve $(A - (a+b)I)\mathbf{v} = \mathbf{0}$:

$$\begin{pmatrix} -b & b & 0 \\ b & -b & 0 \\ 0 & 0 & c-a-b \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

This gives $-bx + by = 0$, so $x = y$ (since $b \neq 0$). Also $(c-a-b)z = 0$. If $c \neq a+b$, then $z = 0$. Choosing $x = 1$ gives $y = 1$. Thus $\mathbf{v}_3 = (1,1,0)^T$.

Special cases: If $c = a - b$, then $\lambda_1 = \lambda_2$ (repeated eigenvalue). If $c = a + b$, then $\lambda_1 = \lambda_3$ (repeated eigenvalue). In these cases, the eigenspace has dimension 2.

(d) The matrix A is diagonalizable if and only if it has three linearly independent eigenvectors.

Case 1: If $c \neq a+b$ and $c \neq a-b$, then we have three distinct eigenvalues, so A is diagonalizable with:

$$P = \begin{pmatrix} 0 & 1 & 1 \\ 0 & -1 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} c & 0 & 0 \\ 0 & a-b & 0 \\ 0 & 0 & a+b \end{pmatrix}$$

Case 2: If $c = a+b$ or $c = a-b$, then we have a repeated eigenvalue. The matrix is still diagonalizable because the 2×2 upper-left block has distinct eigenvalues when $b \neq 0$, ensuring we still have enough eigenvectors.

Therefore, $A = PDP^{-1}$ where P is the matrix of eigenvectors and D is diagonal.

Question 4 (20 marks)

Consider the function $f(x, y) = x^2 + y^2$ subject to the constraint $g(x, y) = x + 2y - 4 = 0$.

(a) Set up the Lagrangian function $\mathcal{L}(x, y, \lambda)$ for this constrained optimization problem. [3 marks]

(b) Find the critical points by solving the system of equations given by the partial derivatives of the Lagrangian. [5 marks]

(c) Classify the critical points as local minima, maxima, or saddle points. [5 marks]

(d) Verify your result by substituting the critical points into $f(x, y)$ and interpreting the values. [4 marks]

(e) Explain the geometric interpretation of your solution. [3 marks]

(a) The Lagrangian function is:

$$\mathcal{L}(x, y, \lambda) = f(x, y) - \lambda g(x, y) = x^2 + y^2 - \lambda(x + 2y - 4)$$

(b) Setting the partial derivatives equal to zero:

$$\frac{\partial \mathcal{L}}{\partial x} = 2x - \lambda = 0 \Rightarrow x = \frac{\lambda}{2}$$
$$\frac{\partial \mathcal{L}}{\partial y} = 2y - 2\lambda = 0 \Rightarrow y = \lambda$$
$$\frac{\partial \mathcal{L}}{\partial \lambda} = -(x + 2y - 4) = 0 \Rightarrow x + 2y = 4$$

Substituting the first two equations into the constraint:

$$\frac{\lambda}{2} + 2\lambda = 4 \Rightarrow \frac{5\lambda}{2} = 4 \Rightarrow \lambda = \frac{8}{5}$$

Therefore: $x = \frac{4}{5}$ and $y = \frac{8}{5}$.

The critical point is $(x, y) = \left(\frac{4}{5}, \frac{8}{5}\right)$.

(c) To classify the critical point using first principles, we parameterize the constraint and analyze the behavior of f along it.

From the constraint $x + 2y = 4$, we can express $x = 4 - 2y$. Substituting this into $f(x, y) = x^2 + y^2$:

$$\begin{aligned} f(y) &= (4-2y)^2 + y^2 \\ &= 16 - 16y + 4y^2 + y^2 \\ &= 16 - 16y + 5y^2 \end{aligned}$$

To find the critical point, we differentiate with respect to y:

$$f'(y) = -16 + 10y = 0 \Rightarrow y = \frac{8}{5}$$

This gives $x = 4 - 2\left(\frac{8}{5}\right) = \frac{4}{5}$, confirming our critical point.
To classify this critical point, we compute the second derivative:

$$f''(y) = 10 > 0$$

Since $f''(y) > 0$, the function f has a local minimum along the constraint at the critical point $\left(\frac{4}{5}, \frac{8}{5}\right)$.
Alternatively, we can verify this is a global minimum by noting that $f(y) = 5y^2 - 16y + 16$ is a parabola opening upward, so its unique critical point must be a global minimum on the constraint line.

(d) Substituting the critical point into $f(x, y)$:

$$f\left(\frac{4}{5}, \frac{8}{5}\right) = \left(\frac{4}{5}\right)^2 + \left(\frac{8}{5}\right)^2 = \frac{16}{25} + \frac{64}{25} = \frac{80}{25} = \frac{16}{5}$$

This is the minimum value of f on the constraint line $x + 2y = 4$.
(e) Geometrically, we are finding the point on the line $x + 2y = 4$ that is closest to the origin. The level curves of $f(x, y) = x^2 + y^2$ are circles centered at the origin with radius $\sqrt{x^2 + y^2}$.
The minimum occurs where the constraint line is tangent to one of these circles. At this tangent point, the gradient of f (which points radially outward from the origin) is parallel to the gradient of g (which is the normal to the constraint line). This is precisely the condition $\nabla f = \lambda \nabla g$ from the Lagrange multiplier method.

The point $\left(\frac{4}{5}, \frac{8}{5}\right)$ is the closest point on the line to the origin, at distance $\sqrt{\frac{16}{5}} = \frac{4}{\sqrt{5}}$.

End of Paper